$$
{}^{V[k]}\hat{\mathbf{A}}[k] = \hat{\mathbf{V}}^T[k]\,\hat{\mathbf{A}}[k]\,\hat{\mathbf{V}}[k] \approx \operatorname{diag}\big(\hat{a}_{11}[k],\ \hat{a}_{22}[k],\ \hat{a}_{33}[k]\big)
$$

$$
\begin{aligned}
{}^{V[k]}\bar{\mathbf{A}}[k+1] &= \hat{\mathbf{V}}^T[k]\,\bar{\mathbf{A}}[k+1]\,\hat{\mathbf{V}}[k] \\
&\approx \hat{\mathbf{V}}^T[k]\,\hat{\mathbf{V}}[k+1]\,\operatorname{diag}\big(\hat{a}_{11}[k+1],\ \hat{a}_{22}[k+1],\ \hat{a}_{33}[k+1]\big)\,\hat{\mathbf{V}}^T[k+1]\,\hat{\mathbf{V}}[k]
\end{aligned}
$$

$$
\hat{\mathbf{V}}^T[k]\,\hat{\mathbf{V}}[k+1] \approx \begin{bmatrix} 1 & -\delta\hat{\theta}_w[k] & \delta\hat{\theta}_v[k] \\ \delta\hat{\theta}_w[k] & 1 & -\delta\hat{\theta}_u[k] \\ -\delta\hat{\theta}_v[k] & \delta\hat{\theta}_u[k] & 1 \end{bmatrix}
$$

$$
{}^{V[k]}\bar{\mathbf{A}}[k+1] \approx \begin{bmatrix} \hat{a}_{11}[k+1] & \big(\hat{a}_{11}[k]-\hat{a}_{22}[k]\big)\,\delta\hat{\theta}_w[k] & -\big(\hat{a}_{11}[k]-\hat{a}_{33}[k]\big)\,\delta\hat{\theta}_v[k] \\ \big(\hat{a}_{11}[k]-\hat{a}_{22}[k]\big)\,\delta\hat{\theta}_w[k] & \hat{a}_{22}[k+1] & \big(\hat{a}_{22}[k]-\hat{a}_{33}[k]\big)\,\delta\hat{\theta}_u[k] \\ -\big(\hat{a}_{11}[k]-\hat{a}_{33}[k]\big)\,\delta\hat{\theta}_v[k] & \big(\hat{a}_{22}[k]-\hat{a}_{33}[k]\big)\,\delta\hat{\theta}_u[k] & \hat{a}_{33}[k+1] \end{bmatrix}
$$

$$
\hat{a}_{11}[k] = \hat{a}_{22}[k]
$$

$$
\begin{aligned}
{}^{V[k]}\bar{\mathbf{A}}[k+1] &\approx \begin{bmatrix} \hat{a}_{11}[k+1] & 0 & -\big(\hat{a}_{11}[k]-\hat{a}_{33}[k]\big)\,\delta\hat{\theta}_v[k] \\ 0 & \hat{a}_{22}[k+1] & \big(\hat{a}_{22}[k]-\hat{a}_{33}[k]\big)\,\delta\hat{\theta}_u[k] \\ -\big(\hat{a}_{11}[k]-\hat{a}_{33}[k]\big)\,\delta\hat{\theta}_v[k] & \big(\hat{a}_{22}[k]-\hat{a}_{33}[k]\big)\,\delta\hat{\theta}_u[k] & \hat{a}_{33}[k+1] \end{bmatrix} \\
&= \begin{bmatrix} \hat{a}_{11}[k+1] & 0 & \hat{a}_{13}[k+1] \\ 0 & \hat{a}_{22}[k+1] & \hat{a}_{23}[k+1] \\ \hat{a}_{13}[k+1] & \hat{a}_{23}[k+1] & \hat{a}_{33}[k+1] \end{bmatrix}
\end{aligned}
$$

$$
\delta\hat{\theta}_u[k] = \frac{\hat{a}_{23}[k+1]}{\hat{a}_{22}[k]-\hat{a}_{33}[k]}
$$

$$
\delta\hat{\theta}_v[k] = -\frac{\hat{a}_{13}[k+1]}{\hat{a}_{11}[k]-\hat{a}_{33}[k]}
$$

$$\hat{\mathbf{V}}[k] = \mathbf{R}_y(\hat{\theta}_y[k])\,\mathbf{R}_x(\hat{\theta}_x[k])$$

$$= \begin{bmatrix} \cos\hat{\theta}_y[k] & \sin\hat{\theta}_y[k]\,\sin\hat{\theta}_x[k] & \sin\hat{\theta}_y[k]\,\cos\hat{\theta}_x[k] \\ 0 & \cos\hat{\theta}_x[k] & -\sin\hat{\theta}_x[k] \\ -\sin\hat{\theta}_y[k] & \cos\hat{\theta}_y[k]\,\sin\hat{\theta}_x[k] & \cos\hat{\theta}_y[k]\,\cos\hat{\theta}_x[k] \end{bmatrix}$$

$$\hat{\mathbf{V}}^T[k]\,\hat{\mathbf{V}}[k+1] \approx \begin{bmatrix} 1 & -\delta\hat{\theta}_w[k] & \delta\hat{\theta}_v[k] \\ \delta\hat{\theta}_w[k] & 1 & -\delta\hat{\theta}_u[k] \\ -\delta\hat{\theta}_v[k] & \delta\hat{\theta}_u[k] & 1 \end{bmatrix}$$

$$\hat{\mathbf{V}}[k+1] \approx \hat{\mathbf{V}}[k]\,\hat{\mathbf{V}}^T[k]\,\hat{\mathbf{V}}[k+1]$$

$$= \begin{bmatrix} \cos\hat{\theta}_y[k] & \sin\hat{\theta}_y[k]\,\sin\hat{\theta}_x[k] & \sin\hat{\theta}_y[k]\,\cos\hat{\theta}_x[k] \\ +\sin\hat{\theta}_y[k]\,\sin\hat{\theta}_x[k]\,\delta\hat{\theta}_w[k] & -\cos\hat{\theta}_y[k]\,\delta\hat{\theta}_w[k] & +\cos\hat{\theta}_y[k]\,\delta\hat{\theta}_v[k] \\ -\sin\hat{\theta}_y[k]\,\cos\hat{\theta}_x[k]\,\delta\hat{\theta}_v[k] & +\sin\hat{\theta}_y[k]\,\cos\hat{\theta}_x[k]\,\delta\hat{\theta}_u[k] & -\sin\hat{\theta}_y[k]\,\sin\hat{\theta}_x[k]\,\delta\hat{\theta}_u[k] \\ \cos\hat{\theta}_x[k]\,\delta\hat{\theta}_w[k] & \cos\hat{\theta}_x[k] & -\sin\hat{\theta}_x[k] \\ +\sin\hat{\theta}_x[k]\,\delta\hat{\theta}_v[k] & -\sin\hat{\theta}_x[k]\,\delta\hat{\theta}_u[k] & -\cos\hat{\theta}_x[k]\,\delta\hat{\theta}_u[k] \\ \\ -\sin\hat{\theta}_y[k] & \cos\hat{\theta}_y[k]\,\sin\hat{\theta}_x[k] & \cos\hat{\theta}_y[k]\,\cos\hat{\theta}_x[k] \\ +\cos\hat{\theta}_y[k]\,\sin\hat{\theta}_x[k]\,\delta\hat{\theta}_w[k] & +\sin\hat{\theta}_y[k]\,\delta\hat{\theta}_w[k] & -\sin\hat{\theta}_y[k]\,\delta\hat{\theta}_v[k] \\ -\cos\hat{\theta}_y[k]\,\cos\hat{\theta}_x[k]\,\delta\hat{\theta}_v[k] & +\cos\hat{\theta}_y[k]\,\cos\hat{\theta}_x[k]\,\delta\hat{\theta}_u[k] & -\cos\hat{\theta}_y[k]\,\sin\hat{\theta}_x[k]\,\delta\hat{\theta}_u[k] \end{bmatrix}$$

$$\hat{\mathbf{V}}[k+1] = \begin{bmatrix} \cos\hat{\theta}_y[k+1] & \sin\hat{\theta}_y[k+1]\,\sin\hat{\theta}_x[k+1] & \sin\hat{\theta}_y[k+1]\,\cos\hat{\theta}_x[k+1] \\ 0 & \cos\hat{\theta}_x[k+1] & -\sin\hat{\theta}_x[k+1] \\ -\sin\hat{\theta}_y[k+1] & \cos\hat{\theta}_y[k+1]\,\sin\hat{\theta}_x[k+1] & \cos\hat{\theta}_y[k+1]\,\cos\hat{\theta}_x[k+1] \end{bmatrix}$$

$$0 = \cos\hat{\theta}_x[k]\,\delta\hat{\theta}_w[k] + \sin\hat{\theta}_x[k]\,\delta\hat{\theta}_v[k]$$

$$\delta\hat{\theta}_w[k] = -\tan\hat{\theta}_x[k]\,\delta\hat{\theta}_v[k]$$

The limit at $\hat{\theta}_x = 90^\circ$ comes from defining u from x: at $\hat{\theta}_x = 90^\circ$ the 0 in the second row of $\hat{\mathbf{V}}$ holds for every $\hat{\theta}_y$, so u is not unique.

$$
\begin{aligned}
-\sin\hat{\theta}_x[k+1] &= -\sin\hat{\theta}_x[k] - \cos\hat{\theta}_x[k]\,\delta\hat{\theta}_u[k] \\
&\approx -\sin\left(\hat{\theta}_x[k] + \delta\hat{\theta}_u[k]\right) \\
\hat{\theta}_x[k+1] &= \hat{\theta}_x[k] + \delta\hat{\theta}_u[k]
\end{aligned}
$$

$$
\begin{aligned}
-\sin\hat{\theta}_y[k+1] &= -\sin\hat{\theta}_y[k] + \cos\hat{\theta}_y[k]\ \sin\hat{\theta}_x[k]\,\delta\hat{\theta}_w[k] - \cos\hat{\theta}_y[k]\ \cos\hat{\theta}_x[k]\,\delta\hat{\theta}_v[k] \\
&= -\sin\hat{\theta}_y[k] - \frac{\cos\hat{\theta}_y[k]}{\cos\hat{\theta}_x[k]}\,\delta\hat{\theta}_v[k] \\
&\approx -\sin\left(\hat{\theta}_y[k] + \frac{\delta\hat{\theta}_v[k]}{\cos\hat{\theta}_x[k]}\right) \\
\hat{\theta}_y[k+1] &= \hat{\theta}_y[k] + \frac{\delta\hat{\theta}_v[k]}{\cos\hat{\theta}_x[k]}
\end{aligned}
$$